

Weather Forecast Errors and the Distribution of Natural Gas Price Variance: Evidence from TTF Futures

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Abstract

This study investigates whether the three-day-ahead temperature forecast errors are associated with subsequent realised variance in Dutch TTF natural gas futures, and whether this association varies across the conditional distribution of variance. Using data from 2022 to 2025, a Heterogeneous Autoregressive (HAR-RV) specification is estimated via ordinary least squares to assess the conditional mean relationship, controlling for volatility persistence, gas storage levels, heating demand, geopolitical risk, and seasonality. The analysis is extended using quantile regression to examine whether the relationship differs across the conditional distribution. The OLS results indicate that temperature forecast errors exhibit only a weak, statistically insignificant association with the conditional mean of subsequent realised variance.

However, quantile regression reveals a positive and statistically significant association at the upper quantiles, particularly at the 90th percentile, suggesting that the relationship between unexpected temperature information and natural gas variance is concentrated in the upper tail of the distribution. This pattern is broadly consistent across both the full sample and a post-crisis subsample excluding the 2022 European energy crisis, providing preliminary evidence that the result is not solely driven by the exceptional conditions of the 2022 European energy crisis. The findings suggest that the relationship between meteorological forecast uncertainty and natural gas market risk is primarily a tail phenomenon that may be obscured by conditional-mean models.

1 Introduction

Natural gas markets have experienced significant volatility since the 2022 energy crisis, when supply and demand constraints combined to produce extreme price swings, particularly in European natural gas markets. Weather-related uncertainty is a particularly salient consideration for natural gas markets, given the sensitivity of heating demand to temperature fluctuations. Weather has long been established as a driver of natural gas demand, operating through heating and cooling requirements, and a substantial body of literature has identified that weather conditions and storage dynamics are related to natural gas volatility.

However, as detailed in the following literature review, existing research has focused mainly on realised weather outcomes, historical anomalies, degree-day measures, and observed weather variables rather than on the unexpected component of weather information available to market participants beforehand. A weather outcome that closely matches what was forecast conveys comparatively little new information to the market, whereas a forecast error, defined as the discrepancy between the realised and forecasted outcome, more directly captures market participants' surprise at the realised outcome. This distinction between weather levels and weather surprises has received comparatively little attention within the natural gas volatility literature, despite its potential relevance for describing how markets and their participants respond to new information.

This paper examines whether temperature forecast errors are associated with the subsequent variance of Dutch TTF natural gas futures, and whether the strength of this association varies across the conditional distribution of that variance rather than being confined to its conditional mean.

To address this question, a sequential empirical framework is employed. A baseline Heterogeneous Autoregressive (HAR-RV) specification is first estimated via ordinary least squares (OLS) to assess the relationship between temperature forecast errors and the conditional mean of forward natural gas variance, controlling for volatility persistence, storage levels, heating demand, geopolitical risk, and seasonality. The analysis is then extended using quantile regression to examine whether this relationship differs across lower, central, and upper regions of the conditional distribution, motivated by evidence elsewhere in the literature that the relevance of fundamental information to natural gas volatility need not be uniform across market conditions. The robustness of any distributional relationship is further assessed by comparing results across the full sample period and a post-crisis subsample, to determine whether the relationship reflects a stable market feature rather than an artefact of the exceptional market conditions observed in 2022.

The results indicate that temperature forecast errors exhibit only a weak, statistically insignificant association with the conditional mean of forward natural gas variance. However, a stronger and statistically significant positive association is found at the upper quantiles of the conditional distribution, and this pattern is found to be broadly consistent across both the full sample and the post-crisis subsample, subject to the limitations discussed in Section 5.

This study aims to contribute to the existing literature in three respects. First, it offers a measurement contribution by employing forecast errors, rather than realised weather anomalies or degree-day measures, as a proxy for the unexpected component of weather information, distinguishing the effect of weather surprises from the effect of weather levels. Second, it provides a market-context contribution by examining this relationship within the European TTF natural gas market, complementing evidence drawn primarily from North American natural gas markets in the existing literature. Third, it offers a distributional contribution by employing quantile regression to assess whether the association between forecast errors and natural gas variance varies across the conditional distribution, complementing conditional-mean approaches that may otherwise obscure a relationship concentrated in specific regions of that distribution.

2 Literature Review

2.1 Weather, Storage, and Natural Gas Return Volatility

Weather is an important determinant of natural gas demand, particularly through its effect on space-heating requirements. Residential and commercial gas consumption is strongly related to temperature, with heating demand typically increasing as temperatures fall below a specified threshold (R. S. Bower and N. L. Bower 1985; Considine 2000). This relationship is commonly represented using heating degree days (HDD), which measure the extent to which temperatures fall below a reference level and therefore provide an empirical measure of heating requirements (Considine 2000). Colder-than-average conditions increase HDDs and expected heating demand, creating upward pressure on natural gas consumption and, where supply cannot adjust immediately, potentially on prices and return volatility.

The relationship between weather and natural gas volatility has been examined directly by Mu (2007). Using US natural gas futures data, Mu incorporates weather shocks into a GARCH framework, measuring weather shocks as deviations of realised degree days from their historical norms. These variables are included in both the conditional mean and conditional variance equations. The results indicate that weather shocks significantly affect natural gas futures returns and volatility. Importantly, incorporating weather and storage surprises into the model reduces estimated volatility persistence by approximately 40 percent. This suggests that some of the persistence attributed to the internal dynamics of natural gas prices may instead reflect identifiable fundamental information.

Subsequent research provides further evidence that weather conditions contain information relevant to natural gas volatility. Peng et al. (2023), for example, find that temperature, precipitation and solar radiation significantly influence natural gas volatility, while Liang et al. (2022) show that models incorporating weather information, particularly indicators of extreme weather, can improve out-of-sample volatility forecasts. Taken together, these studies indicate that the relationship between weather and natural gas volatility extends beyond the mechanical effect of temperature on consumption. Weather variables can also provide information about expected changes in demand and, consequently, about the future balance between gas supply and demand.

However, the studies differ substantially in how weather information is defined. Mu (2007) identifies weather shocks through deviations from historical norms, Peng et al. (2023) incorporate observed weather variables, and Liang et al. (2022) examine the contribution of weather indicators to volatility forecasting. These approaches demonstrate that weather-related variables are relevant to natural gas volatility, but a historical weather anomaly and a forecast error represent deviations relative to different information benchmarks. A cold day may represent a large deviation from its historical average while generating relatively little additional information for market participants if it was accurately anticipated. Conversely, a relatively ordinary temperature may differ substantially from the forecast that was available beforehand. This distinction motivates the focus of the present study on forecast errors rather than solely on realised weather anomalies.

The response of natural gas prices to weather-related information also depends on the ability of the market to absorb changes in demand. In the short run, supply, inventories and infrastructure may not adjust instantaneously to unexpected changes in consumption, meaning that demand-related information can be transmitted relatively quickly into prices. Storage therefore provides an important mechanism through which the effects of demand shocks may be moderated. The theory of commodity storage suggests that inventories influence the relationship between current shocks and prices because available stocks can be drawn upon when demand increases or accumulated when supply exceeds current consumption (Deaton and Laroque 1992) (Pindyck 1994).

Natural gas storage is particularly important because the relationship between inventories and volatility is not necessarily constant across market conditions. Ergen and Rizvanoghlu (2016) show that the effect of storage on natural gas futures volatility varies across seasons. Their findings indicate that the conventional relationship whereby lower storage is associated with greater volatility is most evident during the winter withdrawal season, while higher-than-expected storage can itself be associated with greater volatility during the injection season. These findings suggest that relationships between fundamental information and volatility can vary across prevailing market conditions.

This provides a broader connection to the present study, although seasonal heterogeneity should not be interpreted as evidence of quantile heterogeneity. Ergen and Rizvanoghlu (2016) examine differences across seasons, whereas the present study examines whether the association between forecast errors and the squared-return volatility proxy varies across conditional quantiles. The relevance of the existing evidence is therefore that relationships between fundamentals and volatil-

ity need not be homogeneous across all observations, rather than that previous research directly establishes quantile-specific effects. Within the natural gas volatility literature reviewed here, comparatively less attention is given to whether the unexpected component of weather information, measured using forecast errors, is associated with natural gas return volatility.

2.2 Weather Forecast Errors and Energy Markets

Weather forecasts provide market participants with information about future conditions before those conditions are realised. Numerical weather prediction models become less accurate as the forecast horizon increases, meaning that uncertainty surrounding future temperature generally increases with the distance of the forecast horizon (Bauer, Thorpe, and Brunet 2015). This is economically relevant for energy markets because expectations about future weather contribute to expectations about heating demand, electricity consumption, procurement requirements and inventory conditions. Weather information therefore has relevance not only because of the temperature that eventually occurs, but also because of the information available about future conditions before they are realised.

This creates an important distinction between a weather anomaly and a weather forecast error. A historical weather anomaly measures the difference between realised conditions and a long-run climatological benchmark. A forecast error instead measures the difference between realised conditions and a previously issued forecast. These measures therefore represent deviations relative to different information benchmarks. A realised temperature may be substantially different from its historical average but generate relatively little additional information if it was accurately anticipated. Conversely, a relatively normal temperature may differ substantially from the forecast that was available beforehand.

balance. The relevance of forecast errors to energy-market outcomes provides a broader motivation for considering forecast-based rather than purely climatological measures of weather. Weather expectations form part of the information used to assess future energy demand and market conditions. A forecast error therefore provides a way of distinguishing the weather outcome that eventually occurs from the information that had been available beforehand. This distinction is relevant to the present study because the research question concerns whether the unexpected component of weather information is associated with subsequent natural gas return volatility.

The natural gas literature has already established that weather information contributes to volatility, particularly where weather conditions affect expectations concerning demand (Liang et al. 2022; Peng et al. 2023). However, using realised weather variables as explanatory variables does not directly isolate the extent to which the realised outcome differed from information available beforehand. The natural gas volatility literature reviewed here has largely focused on realised weather conditions, weather anomalies or degree-day measures rather than explicitly modelling the forecast error associated with those conditions. The present study therefore uses temperature forecast error as a proxy for meteorological information that was not fully anticipated in the earlier forecast.

This interpretation requires an important qualification. Forecast error should not be regarded as a direct measure of the information shock perceived by market participants. Market participants may use forecasts from multiple numerical weather prediction models, commercial providers, ensemble forecasts or proprietary forecasting systems. Consequently, the error associated with any single forecast model represents an observable proxy for unexpected weather information rather than a complete measure of the market’s information set. The GFS forecast error used in the present study should therefore be interpreted as the discrepancy between realised weather and one previously available forecast, rather than as an exact measure of the information surprise experienced by market participants.

The present study uses three-day-ahead temperature forecasts from the Global Forecast System (GFS) to construct this measure. A three-day horizon provides a forward-looking measure of weather expectations while retaining relevance for near-term energy demand. The precise construction of the forecast-error variable and its timing are discussed in the methodology. Importantly, the forecast error is constructed ex post from the difference between the realised temperature and the earlier forecast. It should therefore be interpreted as a measure of the discrepancy between realised conditions and previously available forecast information, rather than as a variable that market participants necessarily observed in the exact form used in the empirical analysis.

This distinction establishes the first research gap addressed by the present study. Within the literature reviewed here, existing research demonstrates that weather conditions and weather anomalies are associated with natural gas volatility, while comparatively less attention is given to whether the unexpected component of weather information, represented by forecast errors, is

associated with subsequent natural gas return volatility. This question is particularly relevant to the TTF market, where expectations concerning future demand and supply conditions form part of the information underlying price formation.

2.3 Modelling Natural Gas Return Volatility

Natural gas returns exhibit time-varying volatility and volatility clustering, whereby periods of relatively large or small price movements tend to persist over time. This has motivated extensive use of conditional-volatility models in energy and commodity markets. Mu (2007), for example, incorporates weather and storage variables into a GARCH framework, while Ergen and Rizvanoglu (2016) allow the relationship between fundamentals and natural gas futures volatility to vary across seasonal conditions. These studies demonstrate the relevance of modelling volatility in conjunction with observable fundamental information rather than treating volatility dynamics as entirely independent of market fundamentals.

An alternative approach is provided by the Heterogeneous Autoregressive (HAR) framework developed by Corsi (2009). HAR models volatility persistence across different time horizons, typically incorporating daily, weekly and monthly components. The underlying motivation is that volatility dynamics may reflect the activities of market participants operating at different frequencies. Andersen et al. (2001) further demonstrate the value of realised measures constructed from high-frequency returns for estimating latent volatility. Conventional HAR-RV models therefore use intraday returns to construct realised volatility measures.

Where high-frequency data are unavailable, squared daily returns can instead provide a practical, although noisier, measure related to daily volatility. The distinction is important: squared returns are not equivalent to high-frequency realised volatility, and a specification using squared daily returns should therefore not be described as a conventional HAR-RV model. In the present study, squared returns are treated as a proxy for daily return volatility, with the exact construction of the dependent variable and resulting volatility specification addressed in the methodology.

GARCH and HAR-type models provide useful frameworks for modelling the conditional dynamics of volatility, but the research question of the present study also concerns whether the association between weather information and the squared-return volatility proxy differs across its conditional distribution. A conventional mean-based specification summarises the relationship through an average conditional effect, whereas a distributional approach allows the relationship to be examined at different conditional quantiles. This provides a complementary perspective on the volatility proxy rather than implying that conventional conditional-volatility models are inadequate.

This distinction is relevant given the evidence that the relationship between natural gas fundamentals and volatility can depend on prevailing conditions. Ergen and Rizvanoglu (2016), for example, demonstrate that storage effects vary across seasons, while Mu (2007) shows that incorporating fundamental variables can substantially alter estimated volatility persistence. These findings suggest that relationships between fundamental information and volatility may not be constant across observations. A distributional approach can therefore complement conventional volatility modelling by examining whether the association between forecast errors and squared returns differs across the conditional distribution of the squared-return measure.

The present study consequently considers both the conditional mean and conditional distribution of its squared-return measure. Ordinary least squares (OLS) is used to examine the conditional mean relationship, while quantile regression is used to examine whether the estimated association varies across different conditional quantiles. In this setting, the OLS specification addresses whether forecast errors are associated with the conditional mean of squared returns, while the quantile specifications examine whether the strength of this association differs across lower, central and upper portions of the conditional distribution.

2.4 Quantile Regression and Distributional Relationships

Quantile regression, introduced by Koenker and Bassett Jr (1978), provides a framework for estimating conditional relationships at different points of the response distribution. Unlike ordinary least squares, which estimates the conditional mean of the dependent variable, quantile regression estimates conditional quantiles indexed by τ . This allows the relationship between an explanatory variable and the dependent variable to vary across the distribution rather than imposing a common effect.

This property is relevant when the dependent variable is skewed, heavy-tailed or characterised by heterogeneous responses to explanatory variables. Squared returns can exhibit substantial variation, with relatively large observations occurring during periods of substantial price movements.

Distributional approaches can therefore provide information about the relationship between explanatory variables and the squared-return measure that is not captured by a single conditional-mean estimate.

Existing energy-market research also demonstrates the relevance of accounting for distributional characteristics and uncertainty. Berrisch and Ziel (2022), for example, model European natural gas prices using skewed and heavy-tailed distributions and demonstrate the importance of accounting for non-normality and changing distributional characteristics. In the electricity market, Nowotarski and Weron (2018) review probabilistic forecasting approaches and discuss the use of distributional methods for representing uncertainty rather than relying exclusively on point forecasts. These studies provide broader support for considering the distributional characteristics of energy-market variables, although they do not constitute direct evidence for the specific quantile-regression specification used in the present study.

The relevance of quantile regression to the present study is therefore not simply that it provides another estimation technique. It allows the research question to be extended from whether forecast errors are associated with squared returns on average to whether that relationship differs across the conditional distribution of squared returns. For example, a forecast error could have a relatively small association with observations near the centre of the conditional distribution but a larger association with observations in the upper quantiles. Such a finding would indicate that the statistical relationship between unexpected weather information and the squared-return measure is stronger at higher conditional values of squared returns.

Importantly, the interpretation of these quantiles must remain precise. An upper conditional quantile represents observations associated with a higher value of the conditional dependent-variable distribution; it does not necessarily imply that the market is in a distinct psychological or structural "high-volatility regime". Similarly, where squared returns are used as the dependent variable, the estimated quantiles describe the conditional distribution of squared returns rather than directly observed latent volatility. The coefficients should therefore be interpreted as differences in the estimated association between forecast error and the squared-return measure across conditional quantiles.

The literature therefore establishes three related but distinct foundations for the present study. First, natural gas research demonstrates that weather and storage fundamentals are associated with natural gas volatility, but the weather measures employed in the literature reviewed here have largely been based on realised conditions, historical anomalies or degree-day measures. Second, the distinction between weather outcomes and prior forecasts provides a basis for treating forecast errors as observable proxies for the unexpected component of weather information, while recognising that a forecast from a particular model does not represent the complete information set available to market participants. Third, existing volatility research provides evidence on conditional variance dynamics and average relationships, while a distributional approach offers an additional way to examine whether the relationship between weather-related information and the squared-return measure varies across its conditional distribution.

The present study addresses these gaps by examining whether temperature forecast errors are associated with TTF natural gas return volatility and whether this relationship differs across the conditional distribution of the squared-return measure. The contribution can therefore be understood in three dimensions. First, it is a measurement extension, using forecast errors rather than historical weather anomalies to construct an observable proxy for the unexpected component of weather information. Second, it is a market-context extension, examining this relationship in the European TTF natural gas market rather than assuming that findings from other natural gas markets necessarily apply to the European context. Third, it is a distributional extension, examining whether the association between forecast errors and squared returns varies across conditional quantiles rather than estimating only an average conditional relationship.

Taken together, the literature suggests a focused unresolved question: while weather conditions are associated with natural gas markets and volatility, it remains less clear within the literature reviewed here whether unexpected temperature information, represented by forecast errors, is associated with natural gas return volatility and whether the strength of this association varies across the conditional distribution of the squared-return measure. The present study addresses this question using temperature forecast errors in TTF natural gas futures and examining their relationship with squared returns through both conditional-mean and quantile-based analysis.

3 Methodology

3.1 Research Design

This project utilises a sequential empirical framework to analyse the influence of three-day temperature forecast errors on the subsequent realised variance of Dutch TTF natural gas futures.

The analysis starts with the estimation of a baseline Heterogenous Autoregressive model for realised variance (Corsi 2009), which provides a benchmark specification that accounts for the persistence of realised variance and natural gas market factors, including natural gas storage, demand for heating, geopolitical risk, and seasonality. Diagnostic tests are used to assess the suitability of the Ordinary Least Squares (OLS) specification and determine whether other modelling approaches could improve the efficiency of the estimates.

The analysis is extended using quantile regression to test whether temperature forecast errors have an asymmetric impact on the conditional variance distribution. This approach is motivated by previous diagnostic evidence explored in Section 4.1 and theoretical expectations that meteorological uncertainty can disproportionately amplify upper tail variance in turbulent market conditions. The analysis is primarily conducted in-sample to examine the heterogeneous distributional relationship between meteorological uncertainty and natural gas variance rather than as a forecasting exercise.

The robustness of the distributional relationship is evaluated across the full sample period (2022-2025) and a post-crisis subsample (2023-2025), to assess whether the relationship reflects a stable feature of normal market conditions or is driven by the extreme levels of variance associated with the 2022 European energy crisis.

3.2 Data & Variables

3.2.1 Data Sources

The following analysis aggregates data from four independent sources for the construction of the variables used in the model specification. These datasets comprise meteorological data, natural gas futures prices, natural gas storage data, and a geopolitical risk time series. All datasets cover a sample period from 1 January 2022 to 31 December 2025, representing the longest period for which complete calendar years of data are available across all variables, with the datasets merged at a daily frequency.

The Open-Meteo (2026) Historical Forecast and Previous Runs APIs were used to obtain forecasted and realised temperature data used for the construction of the key explanatory variable. The Global Forecast System (GFS) seamless model was selected because it provides the longest continuous archive of historical forecast data available on the platform, while alternative models, such as the European Centre for Medium-Range Weather Forecasts (ECMWF) model, exhibited an artificial regime change arising from historical data limitations. Three-day-ahead forecasts and their corresponding realised daily temperatures were collected for Paris, Amsterdam, Brussels, and Frankfurt over the study period.

The key explanatory variable was derived from daily Dutch Title Transfer Facility (TTF=F) closing prices obtained from Yahoo Finance, which was accessed through the yfinance Python library (Aroussi 2026). The TTF was selected as it is the principal pricing benchmark for the European natural gas market and is thereby widely used in both academic research and industry, making it an appropriate measure for calculating and modelling natural gas variance. The yfinance library was additionally chosen because it provides reliable historical price data that can be accessed programmatically, which supports a transparent and reproducible data collection process.

Daily European aggregate gas storage data were obtained from the Aggregated Gas Storage Inventory (AGSI) database API (Gas Infrastructure Europe 2026). The AGSI database is the official European platform for reporting gas storage inventories and provides consistent daily observations with minimal missing data.

Finally, the Iacoviello and Caldara (2020) Geopolitical Risk (GPR) Index was used to obtain a measure of geopolitical risk level to include as an additional control variable in the variance model. The GPR index was selected for its GPRD measure, which is a daily measure of geopolitical risk that is directly comparable to the daily frequency of the other variables used in the analysis, avoiding the need for temporal aggregation or interpolation..

3.2.2 Variable Construction & Definition

The variable of interest in this study is the temperature forecast error, constructed using three-day-ahead temperature forecasts and the corresponding realised temperatures of Paris, Amsterdam, Brussels, and Frankfurt. Temperature observations were aggregated to a single regional series using equal weights across the four cities, providing a proxy for area-wide heating demand conditions in the European gas market. These cities were selected as representative economic and population centres within the North-West European natural gas market, where temperature fluctuations are expected to have a substantial influence on heating demand and, consequently, natural gas consumption. A three-day forecast horizon was selected because it provides a meaningful level of forecast uncertainty while remaining within a period where temperature expectations are sufficiently informative for market participants. Shorter forecast horizons generally are associated with substantially lower uncertainty and therefore provide limited variation in forecast errors, whereas longer forecast horizons are increasingly affected by forecast noise, reducing the reliability of the signal captured by the error measure. Realised temperatures are obtained from the Open-Meteo Historical Forecast API and represent observed daily mean temperatures. Meanwhile, hourly forecasted temperatures are obtained from the Open-Meteo Previous Runs API using the three-day-ahead forecast parameter. These hourly forecasts are aggregated by issue date to construct mean daily temperatures for each city. The resulting forecasted and realised daily temperatures are then averaged across cities to produce a representative regional temperature measure for the European natural gas market. Cities are weighted equally because the present analysis does not specify a theoretical basis for assigning different weights across cities. Regional realised and forecasted temperatures are calculated as:

$$\bar{T}_t = \frac{1}{4} \sum_c^4 (T_{c,t}), c \in \{Paris, Amsterdam, Brussels, Frankfurt\} \quad (1)$$

Where c indexes Paris, Amsterdam, Brussels, and Frankfurt. Temperature forecast error is calculated as:

$$TFE_t = \bar{T}_t^{realised} - \bar{T}_t^{forecast} \quad (2)$$

Positive errors indicate that realised temperatures were warmer than expected, while negative values indicate that temperatures were colder than expected. The measure therefore captures unexpected weather information rather than temperature levels. Natural gas demand is strongly influenced by heating requirements, meaning unexpected temperature deviations can alter market expectations regarding future consumption, storage conditions, and supply-demand balances. These unexpected changes in market expectations may contribute to increased market uncertainty. Furthermore, the signed forecast error is retained rather than converted into an absolute error, as the direction of the temperature surprise may have different effects across the conditional distribution of variance. An unreported specification using the absolute forecast error was also estimated at the conditional mean as a robustness check; however, as this yielded no meaningful improvement in model fit (see Appendix Table 10), it is not discussed further in the main analysis. This is particularly relevant given the use of quantile regression, which allows the relationship between temperature forecast errors and natural gas return variance to be examined at different points of the variance distribution.

The dependent variable is the logarithm of the mean realised variance of Dutch TTF natural gas returns over the five days following the realisation of a temperature forecast error. Daily TTF returns are calculated as the log change in daily closing prices. The daily realised variance is then proxied by squared daily returns, as the high-frequency intraday price data required to construct conventional realised variance measures were unavailable. The dependent variable is therefore calculated as the logarithm of the mean squared returns of the subsequent five trading days, providing a forward measure of market variance following a temperature forecast error.

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right) \quad (3)$$

$$RV_{t+1:t+5} = \ln \left(\frac{1}{5} \sum_{k=1}^5 r_{t+k}^2 + 10^{-12} \right) \quad (4)$$

Where: $RV_{t+1:t+5}$ is the forward five-day realised variance measure, r_t is the daily log return on day t , P_t is the Dutch TTF price on day t , and t denotes the day on which the temperature forecast is verified against realised temperatures.

Weather-related demand shocks may require time to be fully incorporated into market expectations. Averaging daily realised variance over the subsequent five-day window, during which TTF closing prices are available, captures the delayed response of natural gas market uncertainty to unexpected temperature information while reducing the influence of transitory day-to-day fluctuations in variance. The logarithmic transformation is applied as the squared returns exhibit right skewness and are strictly non-negative.

Gas storage levels ($FULL_t$) are obtained directly from the AGSI database and are measured as the percentage of available storage capacity that is filled. The inclusion of this variable is motivated by the established relationship between storage levels and natural gas market dynamics documented in the literature. Lower storage levels can contribute to greater natural gas price volatility by reducing the market's buffer against unexpected demand and supply shocks (Mu 2007).

Heating degree days (HDD_t) are included to capture weather-driven demand fluctuations. Higher HDD values indicate colder weather conditions and correspond to increased heating demand, which can exert upward pressure on natural gas prices. The relationship between HDD_t and price variance is less clear-cut, although the variable is included to control for demand-side weather effects that may indirectly influence variance. HDD_t is constructed as follows:

$$HDD_t = \max(0, 18 - T_t^{realised}) \quad (5)$$

Geopolitical risk is incorporated using the Geopolitical Risk Daily measure ($GPRD_t$) from the Caldara and Iacoviello GPR index. The index measures the frequency of newspaper articles covering adverse geopolitical events, including wars, terrorist incidents, and international political tensions, and is widely used in the academic literature as a proxy for geopolitical uncertainty. Prior literature identifies geopolitical risk as an important determinant of commodity market uncertainty, with heightened geopolitical tensions typically being associated with elevated price variance. Including this variable controls for exogenous shocks arising from political conflicts, international tensions, and broader geopolitical events that may affect the natural gas market.

Monthly dummy variables are included to account for seasonal patterns in the natural gas market. Eleven binary indicators are constructed for February through December, with January omitted as the reference category to avoid perfect multicollinearity. These fixed effects control for recurring seasonal influences that are not captured by the other explanatory variables, reflecting the pronounced seasonality of natural gas demand and pricing.

To account for volatility persistence, the model incorporates HAR-RV lag variables to capture the persistence of variance across heterogeneous time horizons. The HAR-RV framework is motivated by the well-established empirical finding that variance exhibits long-memory behaviour, whereby current variance is influenced by information accumulated over short-, medium-, and long-term horizons. As high-frequency intraday data are unavailable, realised variance cannot be constructed using the conventional manner. Consequently, the HAR-RV variables are instead derived from squared daily returns, which as proxies for daily variance. Specifically, three lagged variables are included: the one-day lag of squared returns, together with the five-day and twenty-two-day lagged averages of squared returns, representing daily, weekly, and monthly variance dynamics, respectively. The daily, monthly and weekly regressors are:

$$RV_t^{(d)} = \ln(r_t^2) \quad (6)$$

$$RV_t^{(w)} = \ln\left(\frac{1}{5} \sum_{k=0}^4 r_{t-k}^2\right) \quad (7)$$

$$RV_t^{(m)} = \ln\left(\frac{1}{22} \sum_{k=0}^{21} r_{t-k}^2\right) \quad (8)$$

Although squared returns provide a noisier measure of variance than realised variance constructed from intraday data, they are widely used as a practical proxy when higher-frequency observations are unavailable. This modified specification therefore preserves the HAR-RV framework's objective of capturing volatility persistence across multiple time horizons while remaining consistent with the frequency of the available data.

The final dataset was constructed by aggregating the four sources. All datasets had their date variables converted to a consistent daily datetime format to ensure temporal alignment. Weather observations were restricted to dates corresponding to available TTF trading data, while gas storage and GPR observations were merged with the weather data using an as-of merge based on the forecast issue date. This approach matched each forecast with the most recently available storage

and geopolitical risk information, reflecting the information that would have been available at the time the forecast was issued and preventing the use of information that would not have been available at the time of forecast. Following the aggregation, the dataset was sorted chronologically, and observations with missing values were removed. A standardised copy of the dataset was also created by applying a simple global z-score to the continuous explanatory variables. This dataset is used only for condition number diagnostics covered in Section 4.2 to distinguish the effect of scale differences from multicollinearity.

Table 1: Variable Definitions

Variable	Role	Description	Unit
$RV_{t+1:t+5}$	Dependent	Log five-day forward realised variance.	Std. dev. of log returns
TFE_t	Independent	Difference between realised temperature and the three-day-ahead forecast.	°C
HDD_t	Control	Heating degree days, $\max(0, 18 - T_t)$.	°C
$GPRD_t$	Control	Daily Geopolitical Risk Index.	Index
$FULL_t$	Control	Natural gas storage level.	Decimal
$RV_t^{(d)}$	Control	Daily HAR realised variance (1-day lag).	Square of log returns
$RV_t^{(w)}$	Control	Weekly HAR realised variance (5-day average lag).	Square of log returns
$RV_t^{(m)}$	Control	Monthly HAR realised variance (22-day average lag).	Square of log returns

3.3 Model Specification & Estimation

The baseline linear relationship between natural gas forward variance and the lagged temperature forecast errors is estimated via ordinary least squares (OLS). The OLS specification is given by:

$$\begin{aligned}
RV_{t+1:t+5} = & \beta_0 \\
& + \beta_1 RV_t^d + \beta_2 RV_t^w + \beta_3 RV_t^m \\
& + \beta_4 HDD_t + \beta_5 FULL_t + \beta_6 GPRD_t + \sum_{m=2}^{12} (\delta_m M_{m,t}) + \epsilon_t
\end{aligned} \tag{9}$$

Where RV_t^d , RV_t^w , and RV_t^m are the HAR-RV lag regressors; HDD_t is the heating degree day level; $FULL_t$ is the aggregate natural gas storage level; $GPRD_t$ is the geopolitical risk index; and $\sum_{m=2}^{12} (\delta_m M_{m,t})$ represent the contribution of the binary monthly dummy variables.

This baseline linear relationship is extended to examine the average effect of the temperature forecast errors on future realised variance:

$$\begin{aligned}
RV_{t+1:t+5} = & \beta_0 + \beta_1 TFE_t \\
& + \beta_2 RV_t^d + \beta_3 RV_t^w + \beta_4 RV_t^m \\
& + \beta_5 HDD_t + \beta_6 FULL_t + \beta_7 GPRD_t + \sum_{m=2}^{12} (\delta_m M_{m,t}) + \epsilon_t
\end{aligned} \tag{10}$$

Where TFE_t denotes the realised temperature forecast error.

Heteroskedasticity- and autocorrelation-consistent (HAC) standard errors are computed using the Newey–West estimator with a maximum lag of five, motivated by the five-day rolling mean construction of the dependent variable, which may partially contribute to serial correlation observed in Section 4.1. A set of diagnostic tests are then applied to the OLS residuals: the Jarque–Bera test for normality, the Durbin–Watson tests for residual autocorrelation, the Breusch–Pagan test for heteroskedasticity, Ramsey’s RESET test for functional-form misspecification, and variance inflation factors (VIF) to assess multicollinearity among the regressors.

While OLS identifies the conditional mean response, natural gas variance frequently exhibits pronounced asymmetry and fat tails. Consequently, the effect of temperature forecast errors may

differ materially across calm and turbulent market states. Standard symmetric GARCH models, designed to capture conditional variance dynamics, do not directly reveal how a shock affects different points of the conditional distribution of variance. To address these limitations, the mean regression is complemented by the quantile regression framework of Koenker and Bassett Jr (1978). The quantile regression specification is:

$$\begin{aligned}
RV_{t+1:t+5}(\tau|TFE_t) = & \beta_0(\tau) + \beta_1(\tau)TFE_t \\
& + \beta_2(\tau)RV_t^d + \beta_3(\tau)RV_t^w + \beta_4(\tau)RV_t^m \\
& + \beta_5(\tau)HDD_t + \beta_6(\tau)FULL_t + \beta_7(\tau)GPRD_t \\
& + \sum_{m=2}^{12}(\delta_m(\tau)M_{m,t}) + \epsilon_t
\end{aligned} \tag{11}$$

$$\tau \in \{0.10, 0.25, 0.5, 0.75, 0.90\} \tag{12}$$

Where $RV_{t+1:t+5}(\tau|TFE_t)$ is the conditional τ -th quantile of the forward variance given the temperature forecast error, and τ denotes the set of quantiles estimated.

To account for structural changes in European natural gas markets, the quantile model is estimated over two sample periods: the full data range and a restricted window from 1 January 2023 to 31 December 2025.

The quantile framework offers advantages over symmetric variance models, including the ability to capture asymmetric responses of variance to temperature surprises that mean-based, or symmetric GARCH-type specifications cannot capture without additional parametric structure. Second, by allowing the coefficients on the HAR-RV components to differ across quantiles, the model reveals how variance persistence varies with the state of the market; for example, it can indicate whether high-variance regimes are more persistent than low-variance regimes. The coefficient of primary interest, $\beta_1(\tau)$, thus measures how a marginal change in the temperature forecast error shifts the conditional distribution of forward variance, rather than only its central tendency.

The robustness of the quantile regression is assessed by examining the coverage of the 80% prediction interval of in-sample forecasts and the frequency of quantile crossings.

Interval coverage is calculated as the proportion of realised values that fall within the predicted 10th and 90th conditional quantiles:

$$Coverage_{80\%} = \frac{1}{N} \sum_{t=1}^N \mathbf{1} \left(\hat{Q}_t^{0.10} \leq RV_{t+1:t+5} \leq \hat{Q}_t^{0.90} \right) \tag{13}$$

Where $\hat{Q}_t^{0.10}$ and $\hat{Q}_t^{0.90}$ are the predicted 10th and 90th quantiles of forward variance, respectively, and $\mathbf{1}$ is the indicator function that equals 1 when the condition is satisfied and 0 otherwise. The expected coverage for a well-calibrated model is 80%, with deviations indicating potential misspecification or inadequacy in capturing the conditional distribution of variance.

Quantile crossings are assessed by examining whether a predicted lower quantile exceeds its corresponding higher quantile. For each adjacent pair of quantiles, the crossing rate is defined as:

$$CrossingRate_{\tau_i, \tau_j} = \frac{1}{N} \sum_{t=1}^N \mathbf{1} \left(\hat{Q}_t^{\tau_i} > \hat{Q}_t^{\tau_j} \right) \tag{14}$$

Where τ_i and τ_j are adjacent quantiles with $\tau_i < \tau_j$. A crossing rate of zero indicates that no quantile-ordering violations occur in the predictions, while a non-zero crossing rate suggests potential issues with the model's ability to maintain the expected ordering of quantiles across the conditional distribution of variance.

4 Results

4.1 Exploratory Data Analysis

Initial inspection of the dataset reveals that the empirical analysis is conducted using 965 daily observations. As the TTF prices dataset contained the fewest observations among the datasets before merging, the expected common sample size was 1,005 observations. The sample was primarily constrained by the construction of the backward-looking HAR variables and the forward-facing volatility measure. A further 13 days of observations were excluded due to missing temperature forecast data, with these observations concentrated in January 2024.

Table 2 reports the summary statistics for the dependent, independent, and control variables, excluding the HAR and monthly dummy variables. These variables exhibit substantial differences in scale, as reflected by their varying means and standard deviations. Both $GPRD$ and $FULL$ exhibit substantially greater means and standard deviations than the other continuous regressors, reflecting the construction of the $GPRD_t$ index and the $FULL_t$ percentage-based measure. The distributional statistics also indicate that most variables depart from normality, with TFE_t being the only notable exception, exhibiting skewness and excess kurtosis close to the values under a normal distribution. The remaining variables exhibit varying degrees of skewness and kurtosis. The implications of the distributions of the dependent and key explanatory variables are further discussed in relation to Figures 2 and 3.

Table 2: Summary Statistics

Variable	Mean	Std. Dev	Skew	Ex. Kurtosis
$RV_{t+1:t+5}$	-6.631	1.233	0.318	-0.254
TFE_t	0.133	0.710	-0.029	1.621
HDD_t	5.496	4.935	0.492	-0.846
$GPRD_t$	129.448	61.491	1.828	7.434
$FULL_t$	70.019	19.953	-0.529	-0.654

Table 3 presents the results of the Jarque-Bera test, which formally assesses the departures from normality. All the variables examined have p-values < 0.001 suggesting that all measures reject the null hypothesis of normality. The results of the Augmented Dickey-Fuller test are also reported to examine the presence of unit roots and assess the stationarity of the variables. Most variables similarly have p-values < 0.001 and reject the null hypothesis of a unit root, indicating stationarity. Only HDD_t fails to reject the null hypothesis at the 5% significance level, with a p-value of 0.069. Nevertheless, given that HDD_t is included as a control variable and falls only marginally outside the rejection region, this is unlikely to materially affect the interpretation of the primary relationship between $RV_{t+1:t+5}$ and the TFE_t .

Table 3: Jarque-Bera and ADF Test Results

Variable	Jarque-Bera	p-value	ADFuller	p-value
$RV_{t+1:t+5}$	18.826	< 0.001	-3.671	0.005*
TFE_t	105.646	< 0.001	-7.817	< 0.001
HDD_t	67.747	< 0.001	-2.728	0.069
$GPRD_t$	2756.795	< 0.001	-5.616	< 0.001
$FULL_t$	62.077	< 0.001	-5.424	< 0.001

* The ADF test for $RV_{t+1:t+5}$ reports $p < 0.001$ both with and without a deterministic trend.

Figure 1 presents the time series for the dependent variable, $RV_{t+1:t+5}$, and the primary independent variable, TFE_t , over the full sample period. The $RV_{t+1:t+5}$ series exhibits a discernible downward trend, along with a pronounced decline from elevated levels of volatility in 2022, coinciding with the European energy crisis. Nevertheless, the formal Augmented Dickey-Fuller test, conducted both with and without a deterministic trend, indicates that the series does not contain a unit root, suggesting that the forward volatility measure does not exhibit a stochastic trend and is therefore unlikely to contribute to a spurious regression. The marked elevation of volatility during 2022 motivates the study's regime analysis between the full sample and the post-crisis period to assess whether the estimated relationship remains stable or is specific to different regimes.

In contrast, the temperature forecast error series shows no comparable trend or structural shift over the sample period, fluctuating around zero throughout. This is consistent with its construction

as a measure of unexpected deviations of temperature. The stable distributional behaviour across the periods indicates that any change in the estimated relationship between TFE_t and volatility is therefore consistent with variance responding to forecast errors rather than to a shift in the series itself over time.

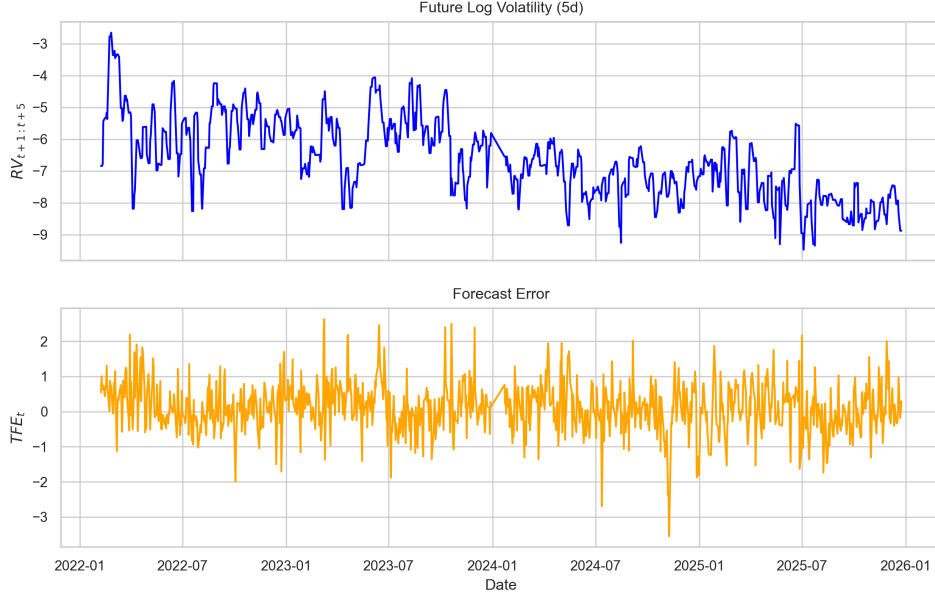


Figure 1: Time Series of $RV_{t+1:t+5}$ and TFE_t

Figure 2 compares the distribution of daily realised variance in both raw and logarithmic form, with corresponding distribution statistics reported in Table 4. The raw realised variance measure exhibits significant right skewness, which is an expected consequence of its construction from squared returns, which enforce strict non-negativity and places a disproportionately greater weight on large price movements. Without transformation, the extreme observations could disproportionately influence the OLS estimation, as both the dependent variable and the HAR-RV lag controls are constructed using the same realised variance measure. In contrast, the logarithmic transformation substantially reduces both skewness and excess kurtosis, from 9.150 to -2.205 and from 113.429 to 12.938, respectively. This results in a more symmetric distribution, as reflected by the substantial decline in the Jarque-Bera statistic from approximately 530,000 to 7,500, which may reduce the influence of extreme observations on the OLS estimation. Nevertheless, the distribution remains statistically non-normal, motivating further analysis of residual normality in Section 4.1.

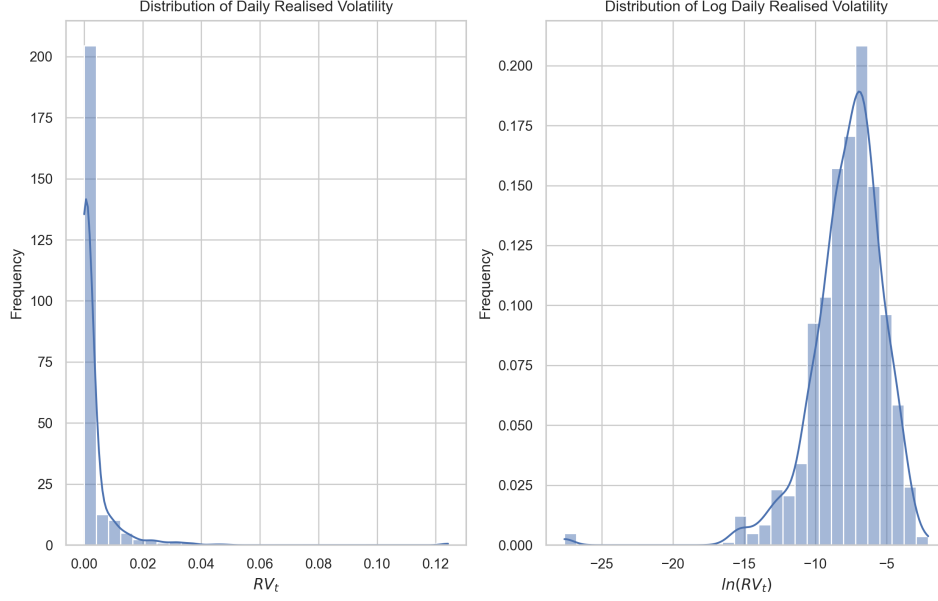


Figure 2: Realised Volatility Distribution

Table 4: Daily Realised Volatility Distributions

Variable	Mean	Std. Dev.	Skewness	Kurtosis	Jarque-Bera	p-value
RV	0.003	0.009	9.150	113.429	530,244.461	< 0.001
$\ln(RV)$	-7.732	2.625	-2.205	12.938	7,505.425	< 0.001

Figure 3 presents the autocorrelation function (ACF) of forward volatility, which highlights a high degree of persistence, with autocorrelation values remaining outside the confidence interval beyond lag 30. Additionally, the partial autocorrelation function (PACF) exhibits several statistically significant spikes beyond the first lag, indicating that the volatility series exhibits significant partial autocorrelation beyond the observations of the preceding period. Therefore, an AR(1) specification may be insufficient to capture the underlying volatility dynamics. This observed persistence motivates the inclusion of the HAR-RV lagged control variables, which provide a parsimonious means of capturing the slow decay in autocorrelation over time. The estimated persistence may be mechanically inflated by the five-day rolling window used to construct the forward volatility measure. A formal decomposition of persistence has not been completed, and therefore the relative contributions of mechanical and genuine persistence has not been determined. Nevertheless, the high degree of autocorrelation in the dependent variable raises the possibility of serial correlation in the regression residuals, which could impact the validity of statistical inferences. Therefore, the residuals of the regression models are tested for serial correlation.

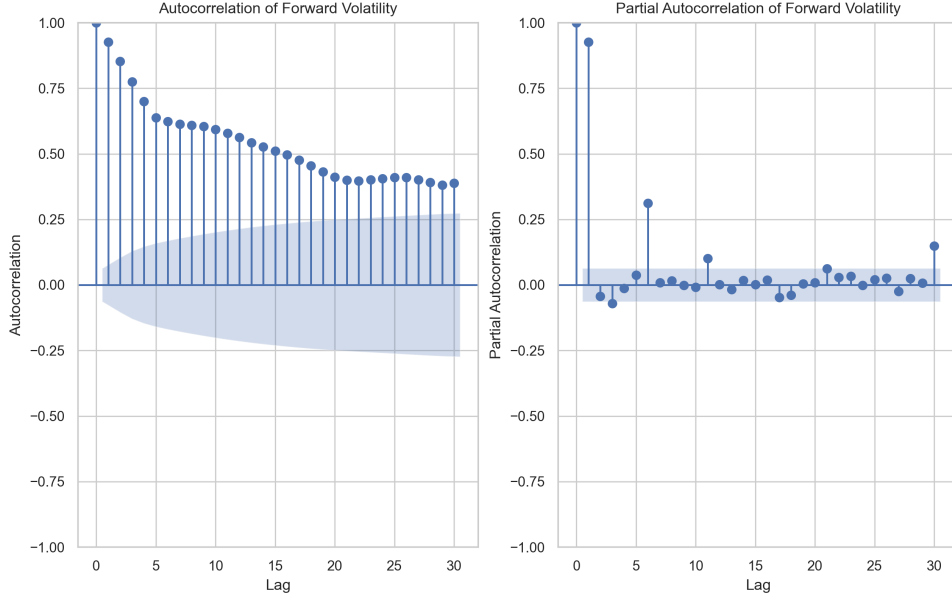


Figure 3: Forward Volatility Autocorrelation & Partial Autocorrelation Functions

Figure 4 presents a hexbin plot of forward realised volatility against temperature forecast error to illustrate their unconditional, univariate relationship. The graph shows density concentrated around $TFE_t = 0$ with no visually apparent directional pattern in the density along either axis, consistent with the weak pairwise correlation reported later in Figure 5. Despite the absence of a simple unconditional association, a genuine relationship may emerge after accounting for other determinants of variance or may be concentrated in specific quantiles of the distribution, which a single linear summary cannot capture. This absence of an unconditional linear relationship is therefore examined using OLS (Section 4.2) and quantile regression (Section 4.3).

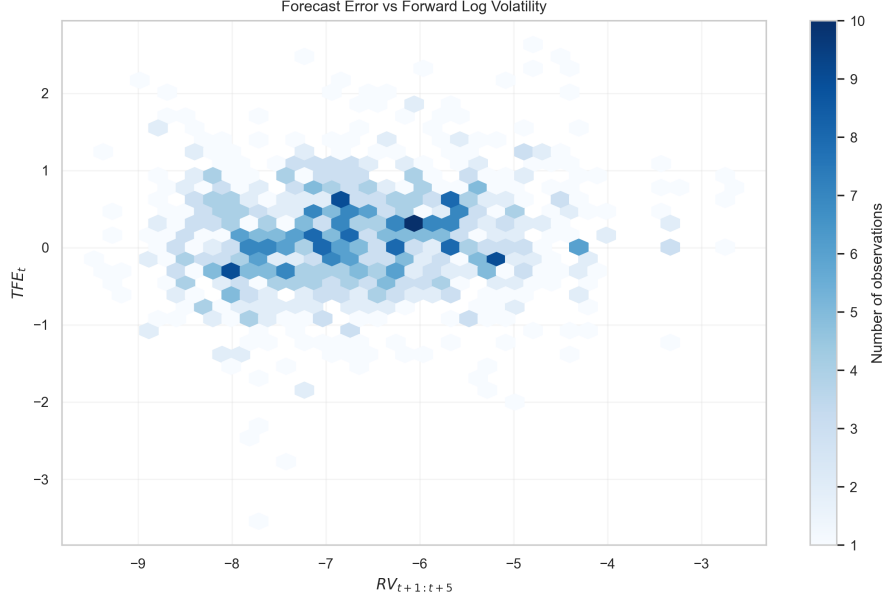


Figure 4: TFE_t vs $RV_{t+1:t+5}$ Hexbin Plot

Figure 5 shows the correlation matrix between all continuous variables present in the regression model specification. In general, most pair-wise correlations are low in magnitude with most having coefficient less than 0.3. However, exceptions occur between the forward variance measure and the weekly and monthly HAR-RV lagged control variables. This is mechanically expected, as they are all rolling averages of the same underlying daily squared-return series. Although this multicollinearity does not bias the regression estimates under the standard assumptions, it can inflate their standard errors. Nevertheless, high but imperfect multicollinearity among control variables is unlikely to materially affect the identification of the key relationship between temperature forecast error and the volatility of natural gas. Despite this, the multicollinearity concern is further examined in Section 4.2 through condition number diagnostics and the variance inflation factor (VIF) testing.

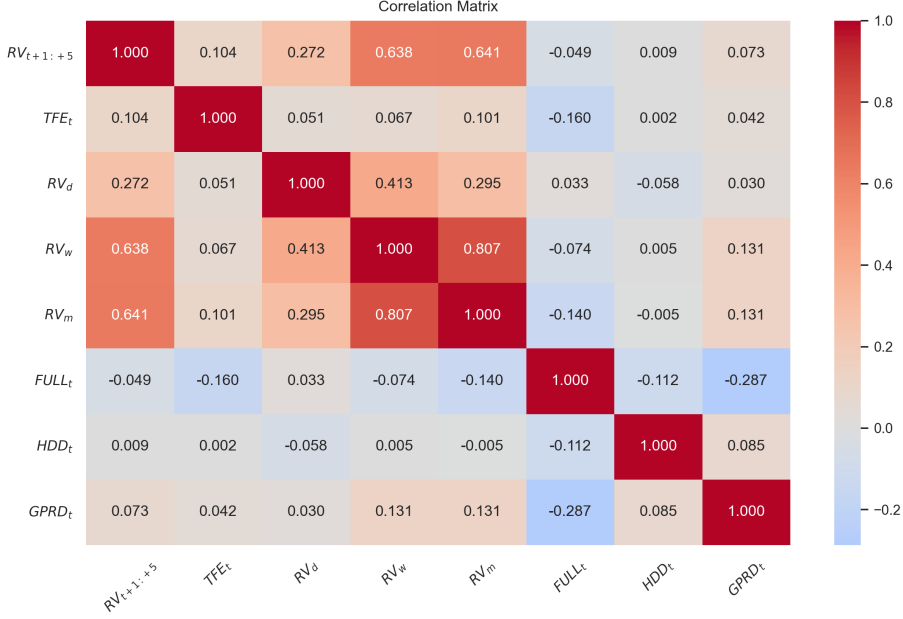


Figure 5: Correlation Matrix of Continuous Variables

Altogether, the exploratory analysis establishes several key features of the underlying dataset. Forward realised volatility is highly persistent and was elevated during the 2022 European energy crisis, while the key explanatory variable, TFE_t , remained relatively stable around zero throughout the sample. The logarithmic transformation of realised variance reduces the influence of extreme observations but still substantially departs from normality. Finally, the presence of some strong correlations between the variables highlights the need to assess potential multicollinearity. These characteristics motivate the inclusion of the HAR-RV lagged volatility controls, the use of logarithmic volatility measures, and the assessment of inference under serial correlation and multicollinearity. Importantly, the EDA does not establish whether temperature forecast errors are systematically associated with subsequent natural gas volatility. Specifically, it does not establish whether the weak unconditional association found reflects a genuine absence of a relationship, an effect that only becomes apparent once other determinants of volatility are controlled for, or an effect that is concentrated within specific quantiles of the distribution of natural gas volatility. The following regression analyses address these possibilities through OLS and quantile regressions.

4.2 OLS Regression

Table 5 presents the results of the OLS regression to test for the presence of a linear association between natural gas volatility and temperature forecast error, conditional on determinants including volatility persistence, heating demand, geopolitical risk, gas storage levels, and seasonality. Table 5 compares the β s and p-values between the baseline (null) specification and the model extended to include TFE_t .

Table 5: OLS Regression Results

Variable	Null Model		TFE_t Model	
	β	p	β	p
const	-2.056	< 0.001	-2.159	< 0.001
TFE_t	–	–	0.091	0.071
RV_t^d	0.007	0.582	0.007	0.607
RV_t^w	0.279	< 0.001	0.282	< 0.001
RV_t^m	0.474	< 0.001	0.468	< 0.001
HDD_t	-0.016	0.405	-0.010	0.603
$GPRD_t$	< 0.001	0.874	< 0.001	0.891
$FULL_t$	0.007	0.287	0.007	0.293
M_2	0.495	0.057	0.499	0.053
M_3	0.356	0.210	0.352	0.207
M_4	-0.315	0.298	-0.311	0.303
M_5	0.155	0.606	0.203	0.506
M_6	0.012	0.972	0.075	0.822
M_7	-0.278	0.395	-0.177	0.594
M_8	0.025	0.943	0.125	0.720
M_9	-0.226	0.443	-0.144	0.628
M_{10}	-0.305	0.306	-0.252	0.401
M_{11}	-0.147	0.555	-0.127	0.609
M_{12}	0.249	0.207	0.262	0.187
R^2	0.484		0.486	
Condition No.*	3060		3100	

* Condition No. for OLS Regression with simple global z-score standardised variables were: Null Model = 27.8; and TFE_t Model = 28.3.

The null model explains 48.4% of the variation in forward realised variance, while including TFE_t increases the proportion of explained variance to 48.6%, an improvement of only 0.2%, indicating that temperature forecast errors contribute negligible additional explanatory power to the conditional mean of the forward variance. The coefficient on TFE_t is positive ($\beta = 0.091$) but only marginally significant at the 10% level ($p = 0.071$) and insignificant at the 5% level. The point estimate suggests a one-degree positive temperature forecast error is associated with a 9.52% ($e^{0.091}$) increase in subsequent 5-day mean volatility holding other factors constant. Nevertheless, given its marginal significance, the estimate should be interpreted cautiously as the evidence for an association between forecast errors and mean forward variance is weak.

Among the control variables, only the weekly (RV_t^w) and monthly (RV_t^m) HAR-RV lags are statistically significant. This is consistent with the slow-decaying autocorrelation of forward variance documented in Figure 3. The daily HAR-RV component (RV_t^d), storage ($FULL_t$), heating degree days (HDD_t), geopolitical risk ($GPRD_t$), and the monthly seasonal dummies are not statistically significant in this specification, suggesting that the medium- and long-horizon volatility persistence accounts for most of the explanatory power of the model, while the remaining exogenous controls add comparatively little.

The condition numbers of approximately 3,000 for both the baseline and extended specification is noteworthy, as they could signal potentially severe multicollinearity. However, further investigation of the OLS regression model, through standardisation of all continuous variables via simple global z-scoring, reveals that the condition number decreases by 99% to below 30. This suggests that the initially high condition number was driven by the substantial difference in scale as documented in Table 2. This interpretation is further reinforced by the VIF results reported in Table 6, which indicate that most variables have low levels of multicollinearity, with only M_8 exhibiting moderate multicollinearity concerns. Although the constant has a high value of 153.17, this is generally not a concern because it is not used as a predictor of forward volatility in the model specification.

Table 6: Variance Inflation Factors (VIF)

Variable	VIF	Interpretation
const*	153.17	High
M_8	5.23	Moderate
M_7	4.94	Acceptable
HDD_t	4.78	Acceptable
M_9	4.41	Acceptable
M_6	4.29	Acceptable
$FULL_t$	4.18	Acceptable
M_{10}	4.13	Acceptable
M_5	4.03	Acceptable
RV_t^w	3.46	Acceptable
M_4	3.33	Acceptable
RV_t^m	3.18	Acceptable
M_3	3.15	Acceptable
M_{11}	2.91	Acceptable
M_2	2.51	Acceptable
M_{12}	2.42	Acceptable
RV_t^d	1.24	Acceptable
$GPRD_t$	1.22	Acceptable
TFE_t	1.14	Acceptable

< 5 = Acceptable; 5 – 10 = Moderate;
> 10 = High.

* High VIF for the constant is generally
not a concern

Having estimated the OLS specification, a series of diagnostic tests was conducted on the model to assess the validity of the OLS assumptions and the adequacy of the model’s functional form, as presented in Table 7.

Table 7: OLS Regression Diagnostics

Test	Statistic	p-value	Interpretation
Jarque-Bera	14.942	< 0.001	Normality Rejected
Ramsey RESET	0.498	0.608	Linearity Not Rejected
Breusch-Pagan	120.189	< 0.001	Homoskedasticity Rejected
Durbin-Watson	0.370	–	Positive Autocorrelation

H_0 : residual normality (Jarque–Bera), correct equation functional form (Ramsey RESET), homoskedastic residuals (Breusch–Pagan), and non-autocorrelated residuals (Durbin–Watson).

The Jarque-Bera value is statistically significant suggesting that the model residuals are not normally distributed consistent with the non-normal distributions of the continuous variables discussed in Section 4.1. Although, normality of residuals is not an OLS assumption meaning that non-normality does not bias OLS coefficients, it affects the validity of exact finite-sample inferences. Nevertheless, given that the regression is estimated using 965 observations, the OLS estimates are approximately normally distributed due to the Central Limit Theorem, and therefore, the non-normality of the residuals is unlikely to significantly affect the results.

The Ramsey Regression Equation Specification Error Test (RESET) is used to diagnose whether the functional form of the regression equation is correctly specified. The insignificant p-value of 0.608 suggests that there is insufficient evidence to reject the null hypothesis. Therefore, there is no evidence of functional-form misspecification or an omitted non-linear variable. Consequently, the functional form of the regression model does not need to be altered.

Homoskedasticity is tested with the Breusch-Pagan test. The significant p-value suggests that there is sufficient evidence to suggest a non-constant variance indicating that the residuals may exhibit heteroskedasticity. Although this heteroskedasticity does not bias OLS estimates, the standard errors become inconsistent estimators for the true variance in finite samples, causing the OLS to no longer be the Best Linear Unbiased Estimator (BLUE), which may potentially cause subsequent significance tests to be unreliable.

The Durbin-Watson statistic of 0.370 is substantially less than two and indicates strong positive first-order autocorrelation in the residuals. Similarly to the impacts of residual heteroskedasticity, the presence of serial correlation does not bias OLS estimates, but it does cause the standard errors

to be inconsistent and therefore unreliable for significance testing.

Figure 6 displays the ACF and PACF to analyse the residuals beyond the first lag. Unlike the ACF of the dependent variable, the lags in the ACF becomes insignificant after the fifth lag; however, the shape of the PACF still closely resembles the forward variance PACF, with a dominant spike at lag-1, and secondary spikes at lags 6, 11 and 30. Overall, while the magnitude of the autocorrelation has been reduced relative to the raw dependent variable series, the continued similarities highlight that the HAR-RV lag specification does not fully account for the serial dependence structure present. Additionally, a sustained negative dip in the residual autocorrelation is observed between lags 18-20, suggesting a cyclical structure not currently captured by the specification; however, no clear mechanism has been identified to explain this pattern.

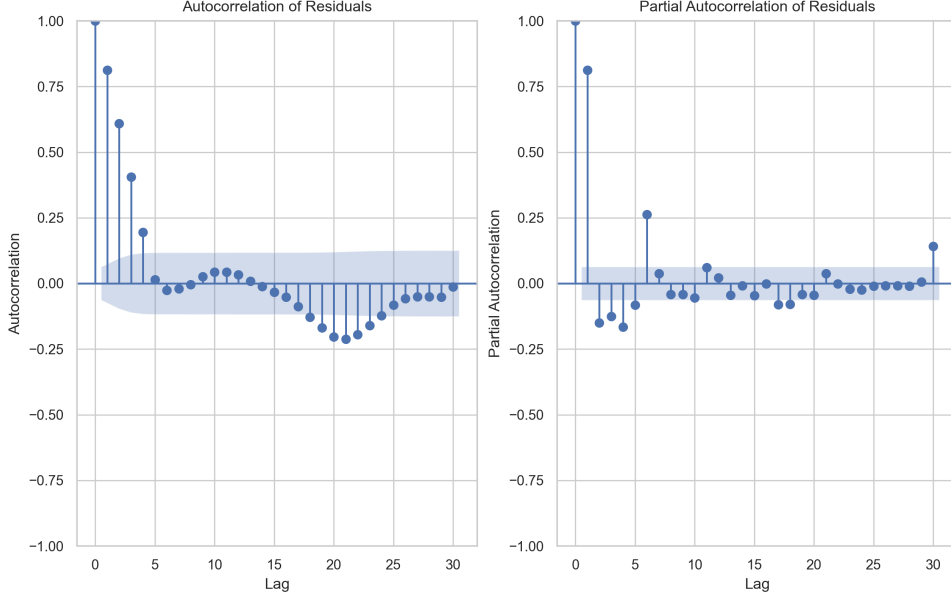


Figure 6: Residual ACF & PACF

The diagnostic tests provide evidence of heteroskedasticity and serial correlation in the regression residuals. Consequently, the standard OLS standard error assumptions are violated. Therefore, the use of HAC standard errors in the OLS regression provides more reliable standard errors and make inference more robust in the presence of the identified heteroskedasticity and serial correlation. On the other hand, the insignificant Ramsey RESET test provides no evidence of functional-form misspecification and therefore the specified relationship between volatility and temperature forecast errors is not rejected by the test. Additionally, although the significant Jarque-Bera test indicates that the residuals are not normally distributed, this is unlikely to materially affect large-sample OLS inference, given the 965 observations and the asymptotic properties of the OLS estimators.

Overall, the OLS results indicate that there is no statistically significant relationship between the temperature forecast error and the conditional mean of subsequent volatility of natural gas futures. Nevertheless, the absence of a statistically significant effect at the mean does not imply that the forecast errors have no effect throughout the distribution of the dependent variable. Therefore, to investigate whether this relationship varies across different quantiles of the conditional distribution, the following section employs quantile regression.

4.3 Quantile Regression

Table 8 reports the quantile regression coefficients and their corresponding p-values, while Figure 7 plots the estimated coefficients across the quantiles, with transparent bands indicating their confidence intervals. The quantile regression results indicate that the association between TFE_t and $RV_{t+1:t+5}$ varies across the conditional distribution. Across both full and post-crisis samples, the coefficients of TFE_t are statistically insignificant at the 10th and 25th quantiles, suggesting limited evidence of a systematic association at the lower quantiles of the conditional distribution. However, the coefficients become more positive and statistically significant towards the upper end

of the distribution. The strongest evidence is observed at the 90th quantile, where the coefficients are highly significant in both samples.

Table 8: Quantile Regression TFE_t Coefficients and p-values

τ	Full Sample		Post-Crisis	
	β	p	β	p
0.10	0.041	0.476	-0.029	0.574
0.25	-0.008	0.852	0.005	0.910
0.50	0.097	0.033	0.079	0.073
0.75	0.069	0.117	0.099	0.015
0.90	0.203	< 0.001	0.149	0.001

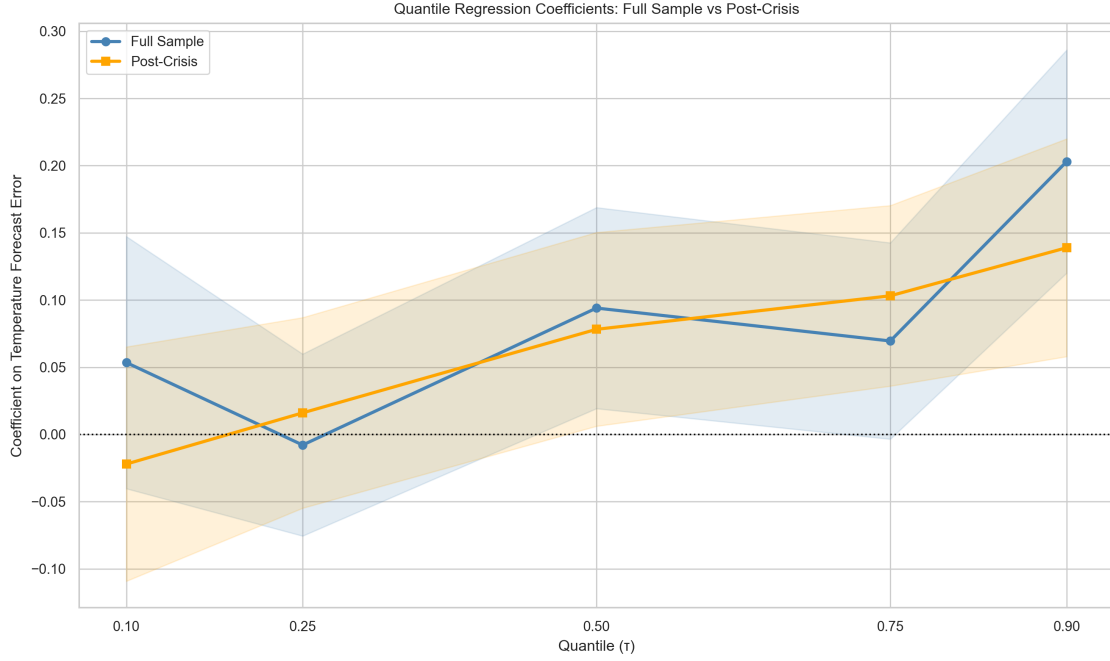


Figure 7: Quantile Regression Coefficients for TFE_t Across Quantiles

Table 9 presents the frequency of quantile crossings across adjacent estimated quantiles. The crossings are concentrated at the extremes of the conditional distribution, with no crossings between the middle quantiles. Crossings between the 0.10 and 0.25 quantiles are significantly more pronounced in the post-crisis subsample, increasing from 1.35% in the full sample to 7.73%. These crossings arise due to the independent estimation of each quantile, which does not impose monotonicity restrictions that higher quantiles must lie above lower ones. This is likely amplified at the tails of the conditional distribution by the relatively small overall sample size, the further reduction of observations in the post-crisis sample, and the greater uncertainty associated with estimating the tails of the conditional distribution, which contributes to the increased uncertainty that can lead to crossing estimates. Overall, these quantile crossing patterns suggest that inferences regarding tail effects should be interpreted with some caution. However, the relatively low frequency of crossings at the upper end suggests that upper-tail estimates are less affected by this issue than those at the lower tail, supporting their interpretation across both samples.

The 80% prediction intervals achieve coverage rates above 78%, which are reasonably close to the nominal level. This indicates that the estimates provide a generally accurate representation of the conditional distribution.

Taken together, the relatively low frequency of crossings at the upper end of the distribution and the near-nominal coverage suggest that the upper tail estimates are reasonably calibrated. However, the frequent crossings between the 0.1 and 0.25 quantiles indicate that greater caution is warranted when interpreting the lower tail estimates.

Table 9: Quantile Prediction Evaluation

Crossing Pairs	Full Sample		Post-Crisis	
	Rate (%)	No.	Rate (%)	No.
0.10-0.25	1.348	13	7.734	57
0.25-0.50	0.000	0	0.000	0
0.50-0.75	0.000	0	0.000	0
0.75-0.90	1.141	11	0.678	5
80% Coverage	78.1		78.4	

5 Discussion

5.1 Main Findings

The results of the empirical analysis provide a nuanced answer to the research question of whether temperature forecast errors are related to TTF natural gas futures’ volatility. At the conditional mean, the empirical results indicate only a weak association between temperature forecast error and natural gas volatility, suggesting that, on average, temperature forecast errors are not a substantial source of variation. However, the quantile results demonstrate a substantially stronger association in the upper-tail of the distribution of natural gas volatility, suggesting that the absence of a meaningful mean relationship does not imply that the association is uniform across the entire distribution. The following discussion therefore considers the implications of the contrast between the mean-based and distributional results, before assessing whether the findings remain evident across alternative sample specifications.

The weak mean-based relationship is indicated by TFE_t ’s statistically insignificant OLS coefficient (Table 5) of 0.091 (p-value = 0.071) that is associated with a negligible incremental R^2 improvement of 0.002, indicating that temperature forecast errors explain only an additional 0.2% of the variation in TTF squared returns. This null result at the conditional mean is unlikely to be attributable to the signed construction of the forecast error variable because it is consistent with the supplementary $|TFE_t|$ specification reported in Appendix Table 10, suggesting that the insignificant result is not an artifact of positive and negative surprises offsetting one another. Instead, a more plausible explanation lies in the highly significant HAR-RV lag terms, which indicate the dominance of volatility persistence in explaining the majority of the variation in subsequent volatility within the specification. Practically, this suggests that temperature forecast errors are not useful for improving average, day-to-day volatility forecasts of TTF futures within a linear, mean-based framework. Nevertheless, this insignificant association with the conditional mean does not rule out a meaningful relationship between temperature forecast errors and natural gas volatility overall.

However, the quantile regression results in Table 8 reveal an asymmetric relationship between temperature forecast error and natural gas volatility, with the magnitude and significance of the effect varying across the conditional distribution. Although the association is statistically insignificant or sensitive to the sample periods at the lower quantiles, the 90th quantile is consistently statistically significant and positive across both the full and post-crisis subsamples. The significance of the results indicates that association between natural gas futures volatility and temperature forecast errors becomes more pronounced during periods of exceptionally high natural gas volatility, rather than exerting a constant influence across the distribution. A possible explanation for this asymmetric relationship is that market participants revising expectations regarding natural gas demand in response to unexpected weather conditions which may have increased sensitivity during periods of elevated market uncertainty or stress. The positive coefficient at the 90th quantile indicates that warmer-than-expected temperatures are associated with higher natural gas futures’ volatility during periods of high volatility. This is somewhat unexpected, as colder-than-expected temperatures would intuitively be expected to place greater pressure on gas demand. One possible explanation may be the role of precautionary gas inventories. Market participants may maintain inventories as a buffer against unexpected changes in demand, reducing the risk of supply shortages and stockouts (Mu 2007; Pindyck 1994). Consequently, colder-than-expected temperatures may be partially absorbed through existing inventory, while warmer-than-expected temperatures may result in lower-than-expected demand and require storage adjustments. Such adjustments may, in turn contribute to increased volatility, particularly during periods of already elevated market uncertainty.

The robustness of the distributional findings is assessed through the alternative sample specifications, quantile crossing rates, and prediction interval coverage. The persistence of the positive and statistically significant coefficient at the 90th quantile across both full and post-crisis samples provides evidence that the association is not solely driven by the atypical crisis period, and therefore may be relatively robust to changes in the sampling period. Further support is provided by the relatively low quantile crossing rates at the upper quantiles, which indicate that the estimated conditional quantiles generally preserve their expected ordering. Additionally, the empirical coverage of the 80% prediction interval is relatively close to its nominal 80% level, suggesting that the estimated intervals are reasonably well calibrated. Overall, these diagnostics provide reassurance on the reliability of the interpretation of the conditional distribution. Although the post-crisis estimates contain approximately 28% fewer observations than the full sample, which contributes to lower statistical power, less precise estimates, and may partly explain the elevated quantile crossing rates observed in the lower tail of the post-crisis specification, the continued significance of the upper tail under the more restrictive specification provides additional evidence in favour of the distributional pattern’s robustness.

Nevertheless, the scope of the empirical analysis limits the interpretation of these findings. Although the results provide evidence of a historical association between temperature forecast errors and TTF natural gas futures’ volatility, the analysis does not establish that the forecast errors themselves have a causal relationship with volatility. Furthermore, the in-sample nature of the analysis only describes the historical relationship between the TFE_t and $RV_{t+1:t+5}$ rather than identifying how the upper tail association translates into improved performance in real-time prediction, which remains an open empirical question that requires out-of-sample evaluation. Finally, the interpretation of statistical significance warrants some caution because the presence of residual autocorrelation identified in section 4.1 and the lack of autocorrelation-robust quantile regression standard errors may understate the true quantile coefficient uncertainty.

Despite these qualifications, the findings have potential practical relevance for risk management in the TTF natural gas futures market. While the in-sample results do not establish that TFE_t can improve real-time risk forecasts, they suggest that temperature forecast errors could potentially be used as an additional indicator when monitoring tail-risk conditions.

5.2 Limitations and Extensions for Future Research

Several limitations should be acknowledged when interpreting the results of the analysis. These relate to the construction of the dependent and independent variables, and the generalisability of the relationship between temperature forecast error and natural gas variance.

The measurement of variance presents a limitation of the data. As variance is inherently unobservable, analyses must use observable proxies in the model specification. In the standard HAR-RV framework, Corsi (2009) constructs realised variance using high-frequency intraday returns, consistent with the findings of Andersen et al. (2001). They demonstrate that realised variance constructed from intraday returns is less affected by measurement noise relative to squared daily returns, which contain considerable noise and therefore provide a less precise proxy for latent variance. Consequently, measurement errors affect both the dependent variable and the lagged HAR-RV variance regressors, as these variables are constructed from the same squared daily return proxy. This may thereby increase the variance of estimated coefficients and potentially obscure an underlying relationship. Nevertheless, squared daily returns were employed because intraday price data for Dutch TTF Natural Gas Futures were unavailable through the yfinance dataset. Further research could therefore address this limitation by obtaining high-frequency intraday return data to construct realised variance measures consistent with the HAR-RV framework. The estimated models can be used to assess whether the robustness of the asymmetric relationship identified in the results persists when using more precise measures of variance.

Another limitation relating to the construction of the dependent variable is the use of a five-day rolling average window. The rolling windows cause overlap across consecutive observations, mechanically inducing serial correlation in the quantile regression residuals. This suggests that the reported standard errors may be too small, and consequently the reported relationship may be incorrectly identified as statistically significant. Although HAC standard errors were employed in the OLS regression, only heteroskedasticity-robust standard errors were utilised in the quantile regression due to the lack of support for HAC standard errors in the statsmodel QuantReg functionality (Seabold and Perktold 2010), these do not fully account for the serial dependence induced by the overlapping window structure, which thereby reduces the level of independent information contained in the sample. This limitation could be addressed by retaining only every fifth observation in the variance construction; however, this would reduce the sample size by approximately

80%, decreasing the number of observations from 965 to 191. The substantial loss of observations would constrain the reliability of estimates at the extreme quantiles even further, thus diminishing statistical power at the tails of the distribution where the key findings and inferences are concentrated. It may also potentially exacerbate quantile-ordering violations, which are already prevalent at the lower tail of the post-crisis samples' conditional distribution despite a comparatively smaller 28% reduction in observations. A natural extension of this analysis would therefore be to replicate the analysis on a larger sample, which would permit non-overlapping windows to be employed without compromising the statistical power available to test the relationship between natural gas variance and temperature forecast error.

The sole use of the GFS model as the source of temperature forecasts to construct the key explanatory variable limits the generalisability of the analysis. The estimated relationship between natural gas variance and temperature forecast error may be specific to the characteristics of the GFS forecast system. Consequently, the relationship may not apply to forecast errors generated by alternative models. This limitation could not be addressed within the present analysis, as other publicly available forecasting models did not provide a sufficiently long sample period that matched the sample used in the analysis. An extension of this study would therefore be to incorporate forecasts from multiple forecasting models, such as the ECMWF model. Verifying whether the relationship persists across alternative forecasts would provide a stronger test of robustness and demonstrate that the asymmetric effect of temperature forecast errors on the distribution of natural gas variance is a general phenomenon rather than a feature specific to the GFS model.

Beyond the choice of forecasting model, the geographical construction of the weather variables underlying the key explanatory variable further limits the generalisability of the observed relationship between temperature forecast error and natural gas variance. Although the four European cities in the construction are economically relevant in terms of size, and thereby their energy and natural gas usage, their selection was not guided by a formal theoretical framework for determining their inclusion or relative importance. Accordingly, equal weighting was adopted. Hence, the estimated relationship may reflect characteristics of the selected locations and may not generalise to other cities and geographical regions. Future research could address these limitations by incorporating a broader set of economically relevant European cities and applying theoretically motivated weighting based on empirical factors such as population or energy consumption. Furthermore, the Dutch TTF natural gas used for the construction of the dependent variable is primarily relevant to European markets. Extending the analysis to comparable natural gas benchmarks, such as Henry Hub in the United States or the Japan Korea Marker in Asia, would provide further evidence of whether the observed relationship extends beyond the European market.

Finally, the analysis is inherently constrained by its exclusively in-sample scope. Both the OLS and quantile regression specifications are estimated and evaluated on the same data. Although this is consistent with the study's aim of characterising the historical distributional relationship, this limits the findings' applicability for constructing a forecasting model. This approach provides an optimistic assessment as the reported estimates reflect the model's ability to describe the observed sample rather than its generalisability to unseen data, potentially resulting in overfitting, whereby the model captures not only the true relationship but also noise specific to the sample. Consequently, the analysis provides no genuine test of predictive ability of temperature forecast errors, and the results should not be interpreted as evidence that this relationship would hold in a real-time forecasting setting. Future research could address this limitation through an out-of-sample evaluation framework, in which the model is re-estimated using only data available up to a given point and evaluated against subsequent, unseen observations. Forecast accuracy could then be assessed using quantile-specific scoring rules, such as the pinball loss function to determine whether the distributional relationship identified in-sample contributes genuine, incremental predictive value for characterising future natural gas variance.

Taken together, these limitations do not undermine the findings of the analysis. However, they indicate that further evidence is required to establish their broader robustness.

6 Conclusion

This study examined whether temperature forecast errors are associated with subsequent volatility in Dutch TTF natural gas futures, motivated by the theoretical expectation that meteorological uncertainty may have a disproportionate effect on natural gas volatility during periods of elevated demand-side sensitivity. A baseline specification was first estimated using OLS, before extending the analysis to quantile regression to assess whether the relationship between temperature forecast errors and volatility differs across the conditional distribution of volatility.

The findings indicate that temperature forecast errors do not meaningfully explain variation of the conditional mean of natural gas futures volatility. However, at higher quantiles of the conditional distribution, a positive and statistically significant association emerges. This result is consistent across both the full sample and the post-crisis subsample, indicating that the finding may reflect a persistent feature of the market rather than attributable solely to the extreme conditions observed during the 2022 European energy crisis. These results demonstrate the value of moving beyond mean-based approaches when characterising the relationship between weather-related uncertainty and commodity market risk, as a relationship that is negligible on average may nonetheless be substantial during periods of heightened uncertainty, precisely when accurate risk assessment matters most.

Notwithstanding the study's limitations, which include the use of squared returns as a proxy for variance, the use of a single forecasting model and a fixed set of European cities, and the absence of out-of-sample evaluation, the findings provide an initial indication that meteorological forecast uncertainty may be relevant for understanding risk dynamics in natural gas markets, and motivates several directions for future analysis. These include the use of higher-frequency intraday returns data to construct realised variance, the incorporation of multiple forecasting models and a broader set of European cities, and the evaluation of the relationship in an out-of-sample forecasting framework. These would help establish whether the asymmetric relationship identified reflects a general and exploitable feature of natural gas markets for volatility forecasting and risk management, or one specific to the sample and data sources examined in this study.

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7 Appendix

7.1 Absolute Temperature Forecast Error OLS Regression Results

Table 10: OLS Regression Results: Absolute Forecast Error Specification

Variable	$ TFE_t $ Model	
	β	p
const	-2.061	0.002
$ TFE_t $	0.006	0.939
RV_t^d	0.007	0.578
RV_t^w	0.279	< 0.001
RV_t^m	0.474	< 0.001
HDD_t	-0.016	0.409
$GPRD_t$	< 0.001	0.874
$FULL_t$	0.007	0.287
M_2	0.496	0.057
M_3	0.355	0.208
M_4	-0.316	0.297
M_5	0.156	0.607
M_6	0.012	0.971
M_7	-0.276	0.399
M_8	0.026	0.939
M_9	-0.225	0.447
M_{10}	-0.304	0.309
M_{11}	-0.147	0.555
M_{12}	0.249	0.206
R^2	0.484	
Condition No.	3070	

* Condition No. for OLS Regression with simple global z-score standardised variables was 27.9 for the $|TFE_t|$ Model.